

CSCI 135 Assignment – Strings and Loops

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September 20th / 22nd, 2023

Due Date: Sunday, 9/24/2023 11:59PM

Points: 30

Topic: Introduction to Strings and Loops

The goal of this assignment is to gain experience using strings and loops. Continue making your code organized and readable. Pay attention to your header comment and include comments in-line with your code. Use proper indentation and understandable variable names. Use whitespace to help with readability and logical grouping of sections of code.

Submission:

After following the instructions below, please submit your Python script in Moodle. Upload this as a standalone file – do not zip it. Be sure the file has a header section including the file name, your name, a description of the program, and other useful information such as example usage, if appropriate. (Even if the program doesn't take command line arguments, you can still show example usage to demonstrate that it doesn't take command line arguments.)

- PickTeams.py

Grading:

This program/script is worth 30 points. You will be graded according to the following criteria:

	Points
Header Comment	5
Program Written Correctly, Following Direction	10
Program Runs Correctly, Including Correct Output	15
Total	30

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PickTeams.py

In this assignment, you will use loops (for and/or while), as well as strings.

You will write a program that allows users to enter team and player information which will subsequently be printed. It will operate as follows:

- The user will enter the number of teams using interactive keyboard input, not via command line arguments. The allowable number of teams is between 2 and 10.
- If the user enters a number that is not between 2 and 10 (inclusive), the program should exit with the message "Please enter between 2 and 10 teams."
- Given the number of teams, prompt the user for team names and the names of players on each team until the user enters done. See example below.
- Include meaningful comments in line with your code, especially with your loops.

Example Run:

```
-----  
This program collects information about teams including team names and player names.  
It creates and displays a report of teams and players at the end.  
-----
```

```
How many teams are there? 2
```

```
Please enter a team name when prompted. Then enter player names for that team.  
When done entering player names for that team, type done.
```

```
Please enter the name of Team #1: 1995-96 Chicago Bulls
```

```
Please enter the name of a player for team 1995-96 Chicago Bulls: Ron Harper
```

```
Please enter the name of a player for team 1995-96 Chicago Bulls: Michael Jordan
```

```
Please enter the name of a player for team 1995-96 Chicago Bulls: Luc Longley
```

```
Please enter the name of a player for team 1995-96 Chicago Bulls: Scottie Pippen
```

```
Please enter the name of a player for team 1995-96 Chicago Bulls: Dennis Rodman
```

```
Please enter the name of a player for team 1995-96 Chicago Bulls: Toni Kukoc
```

```
Please enter the name of a player for team 1995-96 Chicago Bulls: done
```

```
Please enter the name of Team #2: 1975 Cincinnati Reds
```

```
Please enter the name of a player for team 1975 Cincinnati Reds: Johnny Bench
```

```
Please enter the name of a player for team 1975 Cincinnati Reds: Tony Perez
```

```
Please enter the name of a player for team 1975 Cincinnati Reds: Joe Morgan
```

```
Please enter the name of a player for team 1975 Cincinnati Reds: Pete Rose
```

```
Please enter the name of a player for team 1975 Cincinnati Reds: Dave Concepcion
```

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Please enter the name of a player for team 1975 Cincinnati Reds: George Foster
Please enter the name of a player for team 1975 Cincinnati Reds: Cesar Geronimo
Please enter the name of a player for team 1975 Cincinnati Reds: Ken Griffey
Please enter the name of a player for team 1975 Cincinnati Reds: Jack Billingham
Please enter the name of a player for team 1975 Cincinnati Reds: done

Team #1: 1995-96 Chicago Bulls

Player #1: Ron Harper
Player #2: Michael Jordan
Player #3: Luc Longley
Player #4: Scottie Pippen
Player #5: Dennis Rodman
Player #6: Toni Kukoc

Team #2: 1975 Cincinnati Reds

Player #1: Johnny Bench
Player #2: Tony Perez
Player #3: Joe Morgan
Player #4: Pete Rose
Player #5: Dave Concepcion
Player #6: George Foster
Player #7: Cesar Geronimo
Player #8: Ken Griffey
Player #9: Jack Billingham

Q&A

How do I get user input?

The `input()` function shall be used. Interactive input is read as a string, so you may need to convert it to an int if intended as an integer or use it as is if intended to be a string. For example:

```
num = int(input("Please enter a number: "))
```

will get input from the user with the prompt "Please enter a number", convert it to an int, and reference it with the variable `x`.

```
name = input("Please enter a name: ")
```

will create a string variable.

How do I add more (concatenate/append) to a string variable? This example adds a new line to the end of the existing `myString` variable. The prior contents of `myString` remain with the new line added to the end.

```
myString += "\n"
```

How do I get started?

Read the problem carefully.

Write down the steps you need to do.

Make a plan and possibly create a flowchart.

Then start writing code incrementally:

- First, handle getting user input and exiting the program if needed.
- Next you might want to just print the team names on their own lines based on the input.
- Then decide what which step to tackle next.

Always test your code as you go by writing small chunks at a time. Choice of how to handle loops is important as is string handling.

Upload PickTeams.py to Moodle as the third part of your submission for this assignment.

EXTENSIONS

You will not submit the answers to these extensions as part of your assignment submission. You should work through and understand them to learn more and practice. You may be held responsible for the content within, so you really should look through them and try what you can ...

From the Liang book - NOTE: We will **not** use MyProgrammingLab in the book. You should be able to do exercises from Liang without MyProgrammingLab.

1. *Go through Section 3.6. How are the approaches for formatting similar to the f-strings we used in class? How are they different?*
2. *Do exercises 3.9 and 3.11.*
3. *Work through other exercises and examples related to strings in Chapter 3. Do the easy ones first. Try some of the more challenging ones*
4. *5.10, 5.11, 5.29*
5. *Work through other exercises and examples in Chapter 5. Do the easy ones first. Try some of the more challenging ones*

Work through other exercises and examples in Chapter 5. Do the easy ones first. Try some of the more challenging ones

From the *Things to Do* section at the end of Chapter Five of our optional Lubanovic book:

5.1 Capitalize the word starting with m:

```
>>> song = """When an eel grabs your arm,  
... And it causes great harm,  
... That's - a moray!"""
```

5.2 Print each list question with its correctly matching answer, in the form:

Q: *question*

A: *answer*

```
>>> questions = [  
...     "We don't serve strings around here. Are you a string?",  
...     "What is said on Father's Day in the forest?",  
...     "What makes the sound 'Sis! Boom! Bah!'"  
... ]  
>>> answers = [  
...     "An exploding sheep.",  
...     "No, I'm a frayed knot.",  
...     "'Pop!' goes the weasel."  
... ]
```

5.3 Write the following poem by using old-style formatting. Substitute the strings 'roast beef', 'ham', 'head', and 'clam' into this string:

```
My kitty cat likes %s,  
My kitty cat likes %s,  
My kitty cat fell on his %s  
And now thinks he's a %s.
```

5.4 Write a form letter by using new-style formatting. Save the following string as `letter` (you'll use it in the next exercise):

```
Dear {salutation} {name},  
  
Thank you for your letter. We are sorry that our {product}  
{verbed} in your {room}. Please note that it should never  
be used in a {room}, especially near any {animals}.  
  
Send us your receipt and {amount} for shipping and handling.  
We will send you another {product} that, in our tests,  
is {percent}% less likely to have {verbed}.  
  
Thank you for your support.  
  
Sincerely,  
{spokesman}  
{job_title}
```

5.5 Assign values to variable strings named 'salutation', 'name', 'product', 'verbed' (past tense verb), 'room', 'animals', 'percent', 'spokesman', and 'job_title'. Print `letter` with these values, using `letter.format()`.

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5.6 After public polls to name things, a pattern emerged: an English submarine (Boaty McBoatface), an Australian racehorse (Horsey McHorseface), and a Swedish train (Trainy McTrainface). Use % formatting to print the winning name at the state fair for a prize duck, gourd, and spitz.

5.7 Do the same, with `format()` formatting.

5.8 Once more, with feeling, and *f strings*.

From the *Things to Do* section at the end of Chapter Six of our optional Lubanovic book:

6.1 Use a `for` loop to print the values of the list `[3, 2, 1, 0]`.

6.2 Assign the value 7 to the variable `guess_me`, and the value 1 to the variable `number`. Write a `while` loop that compares `number` with `guess_me`. Print 'too low' if `number` is less than `guess_me`. If `number` equals `guess_me`, print 'found it!' and then exit the loop. If `number` is greater than `guess_me`, print 'oops' and then exit the loop. Increment `number` at the end of the loop.

6.3 Assign the value 5 to the variable `guess_me`. Use a `for` loop to iterate a variable called `number` over `range(10)`. If `number` is less than `guess_me`, print 'too low'. If it equals `guess_me`, print 'found it!' and then break out of the `for` loop. If `number` is greater than `guess_me`, print 'oops' and then exit the loop.

See the *Things to Do* section at the end of Chapter Five of our book.

See the *Things to Do* section at the end of Chapter Six of our book.

From me:

1. It is important to know that Python string objects are immutable because of the impact on performance on string operations. What impact(s) can you think of?
2. Python does not have a `do-while` construct where the condition check is done at the end of each iteration, rather than the beginning. How could a `do-while` loop be implemented using the `while` construct in Python?
3. How could use a loop to revise your `Lottery.py` script from the prior lab to ensure that the two lottery numbers do not match?